

HOMEWORK 3
ELEMENTARY LINEAR ALGEBRA
(MATH 2250-03), FALL, 2018

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SECTION 1.8 - INTRODUCTION TO LINEAR TRANSFORMATIONS

1. Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation. Show that if T maps two linearly independent vectors in $\mathbb{R}^n$ onto a linearly dependent set of vectors $\{T(v_1), T(v_2)\}$ in $\mathbb{R}^m$, then the equation $T(x) = 0$ has a nontrivial solution.

Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation, and suppose T maps two linearly independent vectors in $\mathbb{R}^n$ onto a linearly dependent set of vectors in $\mathbb{R}^m$. We desire to show that this implies the equation $T(x) = 0$ has a nontrivial solution.

To do so, let $\{v_1, v_2\}$ be a set of linearly independent vectors in $\mathbb{R}^n$ such that $\{T(v_1), T(v_2)\}$ is linearly dependent in $\mathbb{R}^m$ (note that such a set of vectors must exist by our initial assumption). By the definition of linear dependence, there exists constants $c_1, c_2 \in \mathbb{R}$, at least one not equal to zero, such that

$$c_1 T(v_1) + c_2 T(v_2) = 0.$$

But now, as T is a linear transformation, this tells us that

$$c_1 T(v_1) + c_2 T(v_2) = 0$$

$$T(c_1 v_1) + T(c_2 v_2) = 0$$

$$T(c_1 v_1 + c_2 v_2) = 0.$$

So, as $\{v_1, v_2\}$ is a linearly independent set of vectors in $\mathbb{R}^n$, we may conclude that $c_1 v_1 + c_2 v_2 \neq 0$ by the definition of linear independence. Thus, we have found a vector $v = c_1 v_1 + c_2 v_2$ with $v \neq 0$ such that $T(v) = 0$ – that is, we have found a nontrivial solution to $T(x) = 0$.

Therefore, we may conclude that if T maps two linearly independent vectors in $\mathbb{R}^n$ onto a linearly dependent set of vectors $\{T(v_1), T(v_2)\}$ in $\mathbb{R}^m$, then the equation $T(x) = 0$ has a nontrivial solution.

2. Let $\{u, v\}$ be a set of linearly independent vectors in $\mathbb{R}^3$. The parametric equation of a plane P through u , v , and 0 is $x = su + tv$, where $s, t \in \mathbb{R}$, i.e.

$$P = \{su + tv \in \mathbb{R}^3 : s, t \in \mathbb{R}\}.$$

Suppose $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is some arbitrary linear transformation. What must be true about $T(u)$ and $T(v)$ in order for the image of the plane P to be a plane?

Let $\{u, v\}$ be a set of linearly independent vectors in $\mathbb{R}^3$, and let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be some arbitrary linear transformation. We seek to determine the conditions that $T(u)$ and $T(v)$ must satisfy in order for the image $T(P)$ to be a plane in $\mathbb{R}^3$, where $P = \{su + tv \in \mathbb{R}^3 : s, t \in \mathbb{R}\}$ is the plane through u , v , and 0 .

To that end, note that if $x \in P$ (that is, if x is a vector lying on the plane P), then $x = su + tv$ for some constants $s, t \in \mathbb{R}$. So, observe that

$$\begin{aligned} T(x) &= T(su + tv) \\ &= T(su) + T(tv) \\ &= sT(u) + tT(v). \end{aligned}$$

Hence, we may conclude that the image of P under T is given by

$$T(P) = \{sT(u) + tT(v) : s, t \in \mathbb{R}\}.$$

So, note that if $\{T(u), T(v)\}$ is linearly dependent, then there exists constants $c_1, c_2 \in \mathbb{R}$, at least one constant not equal to zero, such that $c_1 T(u) + c_2 T(v) = 0$. Without loss of generality, let's suppose $c_1 \neq 0$, since we could always have relabeled the constants if this were not the case. Then, we have

$$\begin{aligned} c_1 T(u) + c_2 T(v) &= 0 \\ c_1 T(u) &= -c_2 T(v) \\ T(u) &= -\frac{c_2}{c_1} T(v). \end{aligned}$$

But now, observe that this then says that

$$\begin{aligned} T(P) &= \{sT(u) + tT(v) : s, t \in \mathbb{R}\} \\ &= \{s(-c_2/c_1)T(v) + tT(v) : s, t \in \mathbb{R}\} \\ &= \{(t - s(c_2/c_1))T(v) : s, t \in \mathbb{R}\}. \end{aligned}$$

Substituting $c = t - s(c_2/c_1)$, this then says that

$$T(P) = \{cT(v) : c \in \mathbb{R}\}.$$

That is, if $\{T(u), T(v)\}$ are linearly dependent, then $T(P)$ is *at best* a line in $\mathbb{R}^3$ (which happens when $T(v) \neq 0$) and *at worst* $T(P)$ is just the zero vector (which happens when $T(v) = 0$).

Therefore, we may conclude that $\{T(u), T(v)\}$ must be a set of linearly independent vectors if $T(P)$ is a plane in $\mathbb{R}^3$.

SECTION 1.9 - THE MATRIX OF A LINEAR TRANSFORMATION

3. Let $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$, $v_1 = \begin{bmatrix} -2 \\ 5 \end{bmatrix}$, and $v_2 = \begin{bmatrix} 7 \\ -3 \end{bmatrix}$, and let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a linear transformation that maps x to $x_1 v_1 + x_2 v_2$, i.e.

$$T(x) = x_1 v_1 + x_2 v_2.$$

Find a matrix A such that $T(x) = Ax$ for all $x \in \mathbb{R}^2$.

Let $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$, $v_1 = \begin{bmatrix} -2 \\ 5 \end{bmatrix}$, and $v_2 = \begin{bmatrix} 7 \\ -3 \end{bmatrix}$, and let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a linear transformation that maps x to $x_1 v_1 + x_2 v_2$, i.e.

$$T(x) = x_1 v_1 + x_2 v_2.$$

We seek a matrix A such that $T(x) = Ax$ for all $x \in \mathbb{R}^2$.

To that end, let us notice that if A is a matrix with columns $a_1, a_2 \in \mathbb{R}^2$ (i.e. if $A = \begin{bmatrix} a_1 & a_2 \end{bmatrix}$), then

$$\begin{aligned} Ax &= \begin{bmatrix} a_1 & a_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \\ &= x_1 a_1 + x_2 a_2. \end{aligned}$$

So, since $T(x) = x_1 v_1 + x_2 v_2$, we need only find a matrix $A = \begin{bmatrix} a_1 & a_2 \end{bmatrix}$ such that

$$x_1 v_1 + x_2 v_2 = x_1 a_1 + x_2 a_2.$$

Clearly, $A = \begin{bmatrix} v_1 & v_2 \end{bmatrix}$ is a solution to this. That is, if

$$A = \begin{bmatrix} -2 & 7 \\ 5 & -3 \end{bmatrix}$$

then $T(x) = Ax$ for all $x \in \mathbb{R}^2$.

As a side note, if you're doing a problem like this and you're not sure you got the correct answer, it's actually pretty easy to check your answer – simply verify that you get the same thing out of $T(x)$ as you get out of Ax . That is, we can just calculate:

$$\begin{aligned} T(x) &= x_1 v_1 + x_2 v_2 \\ &= x_1 \begin{bmatrix} -2 \\ 5 \end{bmatrix} + x_2 \begin{bmatrix} 7 \\ -3 \end{bmatrix} \end{aligned} \qquad \begin{aligned} Ax &= \begin{bmatrix} -2 & 7 \\ 5 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \\ &= x_1 \begin{bmatrix} -2 \\ 5 \end{bmatrix} + x_2 \begin{bmatrix} 7 \\ -3 \end{bmatrix} \end{aligned}$$

So, since we do get the same thing, we know that we got the right answer!

4. Suppose $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a linear transformation which maps $\mathbb{R}^n$ onto $\mathbb{R}^m$ (i.e. T is *surjective*). What relationship between n and m must be true?

Suppose $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a linear transformation which maps $\mathbb{R}^n$ onto $\mathbb{R}^m$ (i.e. T is *surjective*). We seek a relationship between n and m .

To that end, let us recall the following theorem given in lecture (Theorem 12 of section 1.8 in your book):

Theorem 1. Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation, and let $A = \begin{bmatrix} T(e_1) & \dots & T(e_n) \end{bmatrix}$ be the standard matrix for T . Then:

- (a) T maps $\mathbb{R}^n$ onto $\mathbb{R}^m$ (i.e. T is *surjective*) if and only if the columns of A span $\mathbb{R}^m$.
- (b) T is one-to-one (i.e. T is *injective*) if and only if the columns of A are linearly independent.

So, as T maps $\mathbb{R}^n$ onto $\mathbb{R}^m$ by assumption, it must be the case that the columns of the matrix $A = \begin{bmatrix} T(e_1) & \dots & T(e_n) \end{bmatrix}$ span $\mathbb{R}^m$. That is, we may conclude that $\text{span}\{T(e_1), \dots, T(e_n)\} = \mathbb{R}^m$. So, by the definition of span, it follows that for every $b \in \mathbb{R}^m$, there exists constants $c_1, \dots, c_n \in \mathbb{R}$ such that

$$c_1 T(e_1) + \dots + c_n T(e_n) = b.$$

So, since

$$\begin{aligned} Ax &= \begin{bmatrix} T(e_1) & \dots & T(e_n) \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \\ &= x_1 T(e_1) + \dots + x_n T(e_n), \end{aligned}$$

this tells us the system $Ax = b$ has a solution for every $b \in \mathbb{R}^m$ (just set $x_1 = c_1$, $x_2 = c_2$, and so on). Since $Ax = b$ has a solution for all $b \in \mathbb{R}^m$ if and only if A has a pivot position in every row (see Theorem 4 of section 1.4 in your book), A must then have m pivot positions (since A is an $n \times m$ matrix and correspondingly has m rows). But, since the number of pivot positions cannot be greater than the number of columns, this tells us that $m \leq n$ must hold.

Therefore, we may conclude that if $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a linear transformation which maps $\mathbb{R}^n$ onto $\mathbb{R}^m$ (i.e. T is *surjective*), then it must be the case that $m \leq n$.

In plain English, this says something that feels (at least, to me) intuitively true and reasonable: A linear transformation that maps a *smaller* space into a *bigger* space *can't hit everything* in the bigger space.

5. Let $S : \mathbb{R}^p \rightarrow \mathbb{R}^n$ be a linear transformation, and let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation. Show that the mapping $F : \mathbb{R}^p \rightarrow \mathbb{R}^m$ given by $F(x) = T(S(x))$ is a linear transformation.

Let $S : \mathbb{R}^p \rightarrow \mathbb{R}^n$ be a linear transformation, and let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation. We seek to show that the mapping $F : \mathbb{R}^p \rightarrow \mathbb{R}^m$ given by $F(x) = T(S(x))$ is a linear transformation.

To that end, let us recall the definition of linearity:

Definition 1. A transformation $T : \mathbb{R}^p \rightarrow \mathbb{R}^m$ is linear if $T(au + bv) = aT(u) + bT(v)$ for all $a, b \in \mathbb{R}$ and all $u, v \in \mathbb{R}^p$.

So, we need only show that for all $a, b \in \mathbb{R}$ and all $u, v \in \mathbb{R}^p$, F satisfies

$$F(au + bv) = aF(u) + bF(v).$$

Let's show that this is true directly by using the linearity of S and T . That is, let $u, v \in \mathbb{R}^p$ be arbitrary vectors and let $a, b \in \mathbb{R}$ be arbitrary constants. Then, we have:

$$\begin{aligned} F(au + bv) &= T(S(au + bv)) \\ &= T(S(au) + S(bv)) \\ &= T(aS(u) + bS(v)) \\ &= T(aS(u)) + T(bS(v)) \\ &= aT(S(u)) + bT(S(v)) \\ &= aF(u) + bF(v). \end{aligned}$$

Hence, as we have shown that $F(au + bv) = aF(u) + bF(v)$ for arbitrary $u, v \in \mathbb{R}^p$ and arbitrary $a, b \in \mathbb{R}$, we may conclude that $F(au + bv) = aF(u) + bF(v)$ holds for all $u, v \in \mathbb{R}^p$ and all $a, b \in \mathbb{R}$. Therefore, we may conclude that F is linear.

That is, said another way, we have shown that the composition of two linear operators is a linear operator.

SECTION 2.1 - MATRIX OPERATIONS

6. Let $A = \begin{bmatrix} 2 & -3 \\ -4 & 6 \end{bmatrix}$, $B = \begin{bmatrix} 8 & 4 \\ 5 & 5 \end{bmatrix}$, and $C = \begin{bmatrix} 5 & -2 \\ 3 & 1 \end{bmatrix}$. Verify that $AB = AC$ despite the fact that $B \neq C$.

Let $A = \begin{bmatrix} 2 & -3 \\ -4 & 6 \end{bmatrix}$, $B = \begin{bmatrix} 8 & 4 \\ 5 & 5 \end{bmatrix}$, and $C = \begin{bmatrix} 5 & -2 \\ 3 & 1 \end{bmatrix}$. We seek to verify that $AB = AC$ despite the fact that $B \neq C$.

Note that this problem can be answered by direct computation. That is, we may compute the following:

$$\begin{aligned} AB &= \begin{bmatrix} 2 & -3 \\ -4 & 6 \end{bmatrix} \begin{bmatrix} 8 & 4 \\ 5 & 5 \end{bmatrix} & AC &= \begin{bmatrix} 2 & -3 \\ -4 & 6 \end{bmatrix} \begin{bmatrix} 5 & -2 \\ 3 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 & -7 \\ -2 & 14 \end{bmatrix} & &= \begin{bmatrix} 1 & -7 \\ -2 & 14 \end{bmatrix} \end{aligned}$$

So, it is indeed the case that $AB = AC$, despite the fact that $B \neq C$.

7. For two vectors $u, v \in \mathbb{R}^n$, we say the 1×1 matrix (i.e. real number) $u^T v$ is the *inner product* (or *dot product*) of u and v , and we denote the inner product of u and v by $u \cdot v$ (i.e. $u \cdot v = u^T v$). Show that every linear transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}$ can be written in the form $T(x) = v \cdot x$ for some $v \in \mathbb{R}^n$.

We seek to show that every linear transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}$ can be written in the form $T(x) = v \cdot x$ for some $v \in \mathbb{R}^n$. To that end, recall from section 1.8 that there exists a matrix A for any linear transformation T such that $T(x) = Ax$ for every $x \in \mathbb{R}^n$, and recall that this matrix is given by $A = \begin{bmatrix} T(e_1) & \dots & T(e_n) \end{bmatrix}$, where e_i is the i^{th} column of the identity matrix.

In our case, note that $T(e_1), \dots, T(e_n) \in \mathbb{R}$, since T is a linear transformation which sends $\mathbb{R}^n$ to $\mathbb{R}$. So, $A = \begin{bmatrix} T(e_1) & \dots & T(e_n) \end{bmatrix}$ is a $1 \times n$ matrix, and A^T is an $n \times 1$ matrix – i.e. a vector in $\mathbb{R}^n$. Thus, using the fact that $(M^T)^T = M$ for any matrix M , we have

$$\begin{aligned} (A^T) \cdot x &= (A^T)^T x \\ &= Ax \\ &= T(x). \end{aligned}$$

Reversing the order of the equalities above, this tells us that $T(x) = (A^T) \cdot x$. Therefore, by setting $v = A^T$, we may conclude that every linear transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}$ is of the form $T(x) = v \cdot x$ for some $v \in \mathbb{R}^n$.

In fact, what we have just shown in this problem is a special case of the famous *Riesz Representation Theorem* in functional analysis, which says that a version of the above holds even in an *infinite dimensional* vector space, provided the vector space is (in a certain sense) sufficiently similar to $\mathbb{R}^n$ (basically, you just need some way of measuring angles). Kind of cool, huh?

8. For two vectors $u, v \in \mathbb{R}^n$, we say the $n \times n$ matrix uv^T is the *outer product* of u and v , and we denote the outer product by $u \otimes v = uv^T$. Let $e_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $e_2 = 01$. Show that every 2×2 matrix M with real entries can be written in the form

$$M = c_1(e_1 \otimes e_1) + c_2(e_1 \otimes e_2) + c_3(e_2 \otimes e_1) + c_4(e_2 \otimes e_2)$$

for some constants $c_1, c_2, c_3, c_4 \in \mathbb{R}$.

(In general, a version of this formula holds for arbitrary $n \times n$ matrices. But we'll only do the 2×2 case here, since the general formula is more cumbersome to deal with.)

We seek to show that every 2×2 matrix M with real entries can be written in the form

$$M = c_1(e_1 \otimes e_1) + c_2(e_1 \otimes e_2) + c_3(e_2 \otimes e_1) + c_4(e_2 \otimes e_2)$$

for some constants $c_1, c_2, c_3, c_4 \in \mathbb{R}$.

To that end, let's proceed with some direct computation:

$$\begin{aligned} c_1(e_1 \otimes e_1) + c_2(e_1 \otimes e_2) + c_3(e_2 \otimes e_1) + c_4(e_2 \otimes e_2) &= c_1(e_1 e_1^T) + c_2(e_1 e_2^T) + c_3(e_2 e_1^T) + c_4(e_2 e_2^T) \\ &= c_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \end{bmatrix} \\ &\quad + c_3 \begin{bmatrix} 0 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \end{bmatrix} + c_4 \begin{bmatrix} 0 \\ 1 \end{bmatrix} \begin{bmatrix} 0 & 1 \end{bmatrix} \\ &= c_1 \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + c_2 \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + c_3 \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + c_4 \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} c_1 & c_2 \\ c_3 & c_4 \end{bmatrix}. \end{aligned}$$

Therefore, we have shown that if $M = \begin{bmatrix} m_1 & m_2 \\ m_3 & m_4 \end{bmatrix}$ is a matrix with real entries, then

$$M = m_1(e_1 \otimes e_1) + m_2(e_1 \otimes e_2) + m_3(e_2 \otimes e_1) + m_4(e_2 \otimes e_2).$$

As a side note, problems 7 and 8 actually tell us something pretty cool: Every linear transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}$ is induced by an inner product, and every transformation $S: \mathbb{R}^n \rightarrow \mathbb{R}^n$ is induced by a linear combination of outer products. This tells us that *every* vector in $\mathbb{R}^n$ can be identified with a linear transformation from $\mathbb{R}^n$ to $\mathbb{R}$, and *every* linear map from $\mathbb{R}^n$ to $\mathbb{R}^n$ can be built (using the outer product) from vectors in $\mathbb{R}^n$.

Said another way, whether you're studying linear functions from $\mathbb{R}^n$ to $\mathbb{R}$ or linear functions from $\mathbb{R}^n$ to $\mathbb{R}^n$, you are, in some sense, *really just studying* $\mathbb{R}^n$.